

ANZCA 2017 List the methods of analgesia that can be used for patients undergoing open liver resection surgery. (30%). Justify your perioperative pain management strategy. (70%)

Sample Answer (for 2nd part of question)

Pre-op:

I would review patient in clinic for pre-op assessment and counselling, which includes

- Assessment of underlying liver disease and co-morbidities (e.g. renal impairment and coagulopathy)
- Optimisation of existing painful conditions, if any, which may include opioids for cancer-related pain.
 - o Prescribed analgesics by primary oncologist/ surgeon should be continued during the peri-op period.
- Counselling of anaesthesia/ analgesia plan

I would also make use of this opportunity to allay patient's concerns and manage patient's expectations, which could possibly improve patient's satisfaction and self-reported pain scores.

Intra-op:

I would employ an opioid-sparing, multimodal pain management strategy to cover for both somatic (open resection) and visceral (visceral tissue manipulation/ dissection/ trauma) pain. Good intra-op analgesia facilitates maintenance of stable haemodynamics during the surgery, and smooth emergence from anaesthesia at the end of surgery.

- Basic analgesia
 - o I would administer **IV Paracetamol 1g** and **IV Parecoxib 40mg** at the start of the surgery if there are no contra-indications (severe hepatic impairment, platelet dysfunction, hepato-renal syndrome, ischemic heart disease).
 - o *This is in accordance to ESRA PROSPECT recommendation for analgesia for open liver resection.*
- Adjunct analgesia
 - o In line with an opioid sparing technique, I would employ the use of adjunct analgesia intra-op as there are *supporting evidence from RCT studies for open abdominal surgery, showing reduction in opioid usage.*
 - **IV Magnesium Sulphate** 30 mg/kg loading dose, followed by 10 mg/kg/h
 - **IV Dexmedetomidine** 0.6 mcg/kg loading dose, followed by 0.2 mcg/kg/h
 - **IV Ketamine** 300 mcg/kg loading dose, followed by 100 mcg/kg/h
 - o I would stop the infusions 30-40min before the end of the surgery (1) to minimise sedation during emergence and (2) to facilitate NMBA reversal (as MgSO4 potentiates NMB).
- Opioids
 - o **TCI Remifentanyl**, an ultra-short acting opioids with organ-independent metabolism may be titrated intra-op to supplement my basic/ adjunct analgesia, especially if patient was already on opioids prior to surgery.
- Regional analgesia
 - o My preference would be for **bilateral subcostal TAP catheters**, to be inserted at the end of surgery. Single bolus 0.2% Ropivacaine 30ml per side, followed by running 5-7ml/h on each side.

- *There is robust evidence for use of TAP block catheter to reduce pain score and post-op opioid consumption, and is included in ESRA PROSPECT recommendations for open liver surgery.*
- While **thoracic epidural** has been demonstrated to be superior to subcostal TAP block/catheter in terms of analgesia, I would avoid an epidural due to potential concerns of (1) post-resection coagulopathy resulting in increased risk of epidural haematoma, (2) post-op hypotension requiring vasopressor/ fluid resuscitation, which may delay recovery and early mobilisation.

Post-op:

My post-op pain management goals would be to provide adequate analgesia to (1) enable effective deep breathing/chest physiotherapy (to prevent post-op atelectasis/pneumonia), (2) early mobilisation (reducing DVT/PE risk), while (3) avoiding complications associated with opioid use (e.g. respiratory depression, nausea/vomiting, drowsiness, constipation)

- Basic analgesia
 - Regular **Paracetamol** and **COX-II inhibitors (Etoricoxib/ Parecoxib)** should be scheduled for the next 5 days.
 - Dose adjustment of Paracetamol in consultation with surgeon may be required if there is impaired liver function post-resection. Monitoring of post-op renal function while on COX-II inhibitors should also be done.
- Regional analgesia
 - I would continue to run 0.2% Ropivacaine 5-7ml/h per side via **subcostal TAP catheters** for the next 3-5 days. This covers the somatic pain (from surgical incision), but does not provide analgesia from visceral pain.
- Opioids
 - In institution with an established pain service for post-op monitoring, my preference is for **intrathecal morphine 300-400 mcg** (administered pre-induction) to cover for visceral pain. *This reduces the total amount of systemic opioids post-op, improves patient satisfaction, and pain score for 24-30 hours.*
 - I would minimise the use of systemic opioids, and limit its usage for breakthrough pain. **PCA fentanyl** would be my preferred choice for breakthrough pain as it has fast onset with no active metabolites.
- No role for adjunct analgesia (gabapentinoids, dexamethasone) in the post-op period based on current evidence.